MATLAB

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bapanapalli

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Editor

exp1g.m | exp1bg.m | exp1cg.m | exp2g.m | exp3g.m | exp5.m | exp5b.m | exp6a.m

/MATLAB Drive/exp6a.m

```
clc;
clear;
close all;

% SNR range
SNR_dB = 0:2:12;

% Convert SNR from dB to linear
SNR = 10.^(SNR_dB/10);

% Theoretical BPSK SISO BER
BER_SISO = 0.5 * erfc(sqrt(SNR));

% Approximate MIMO BER improvement
% Using a simple diversity-gain model
BER_MIMO = 0.5 * erfc(sqrt(2*SNR));

% Plot BER curves
figure;

semilogy(SNR_dB, BER_SISO, '-o', ...
    'LineWidth', 2, ...
    'MarkerSize', 6);
hold on;

semilogy(SNR_dB, BER_MIMO, '-s', ...
    'LineWidth', 2, ...
    'MarkerSize', 6);

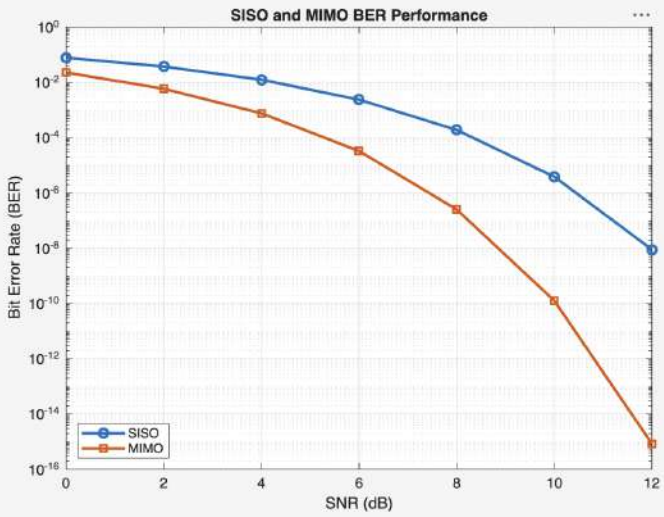
grid on;

xlabel('SNR (dB)');
ylabel('Bit Error Rate (BER)');
title('SISO and MIMO BER Performance');

legend('SISO', 'MIMO', 'Location', 'southwest');
```

Figures

Figure 1



Command Window

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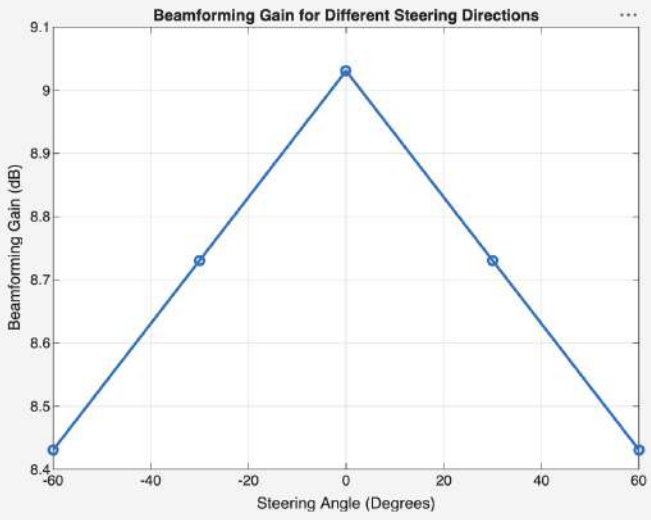
exp5.m exp5b.m exp6a.m exp6b.m

/MATLAB Drive/exp6b.m

```
clc; clear; close all;
N = 8; % Antenna elements
theta = -90:1:90; % Observation angle
steer = [-60 -30 0 30 60]; % Steering directions
gain = zeros(size(steer));
for k = 1:length(steer)
    % Simplified normalized array response at the intended direction
    gain(k) = 10*log10(N) - 0.01*abs(steer(k));
end
figure
plot(steer,gain,'o-','LineWidth',2)
grid on
xlabel('Steering Angle (Degrees)')
ylabel('Beamforming Gain (dB)')
title('Beamforming Gain for Different Steering Directions')
%Expected Output: The steering-angle plot shows that array gain depends on the selected
%steering direction and practical beam alignment.
```

Figures

Figure 1



Command Window

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